

Mechanics

Mechanics: A branch of physical science that deals with energy and forces and their effect on bodies (Webster's)

- Mechanics concerns
 - **Motion** of objects → Velocity and acceleration
 - **Cause** of the motion → Force and energy
- The objects move, but **do not change their properties**
 - Idealized particles and rigid bodies
 - **Mass** and **moment of inertia** are all what matters
- **Newton's Three Laws of Motion**
 - You remember them, right?
 - *Principia* (1687) pretty much wrapped it up

Classical vs. Modern

- “Modern” in physics means “20th century”
 - Quantum Mechanics
 - Relativity
- Classical Mechanics = pre-Quantum Mechanics
 - We include special relativity as well as E&M
- What happened between the 17th and 20th centuries?

Do We Care?

- We know Relativity and QM are the “right answers”
 - Newtonian Mechanics is a human-scale approximation
- Isn't that enough?
 - Why should we learn the theory that has been superseded?

(An advanced course in classical mechanics) introduces no new physical concepts to the graduate student. It does not lead him directly into current physics research. Nor does it aid him, to any appreciable extent, in solving the practical mechanics problems he encounters in the laboratory.

Goldstein, Preface to the First Edition

Why Classical Mechanics?

- Three Good Reasons

- Close connection to Modern Physics
 - Mastering CM gives you clearer view of QM
- Powerful and versatile mathematical tools
 - Indispensable for advanced studies in physics
- Reformulate familiar laws of physics using completely different approaches
 - Cleaner and more general formalism
 - It's just cool

- But first, let's go back to the 17th century...

Newtonian Mechanics

- In principle, **Newton's Equation of Motion** predicted the motion of any object(s) from the force
 - All you need is a big, fast computer
 - **In reality, life was not so easy**
 - Intel was not founded until 1968
 - More fundamentally, the force may not be known
 - It may depend on time, location, or even velocity
 - E.g. ≥ 2 objects attracting each other by gravity
 - Solving 3-body problem turns out to be impossible
- **Quest for more powerful mathematics**

Generalizing Equation of Motion

- Newtonian Mechanics deals with the object's position
 - Goal: finding $x = x(t)$, $y = y(t)$, $z = z(t)$
 - 3 coordinates for each object $\rightarrow 3N$ for N objects
- But there are infinite other ways to describe motion
 - E.g. a more natural way for a pendulum
$$x = L \cos \theta, \quad y = -L \sin \theta, \quad z = 0, \quad \theta = \theta(t)$$
 - Number of free variables may not be $3N$
 - Let's call the new variables **generalized coordinates**
- What are the Equations of Motion for generalized coordinates?

Lagrangian Formulation

- Newton's Equation is about force $\mathbf{F} = m\mathbf{a}$

- You start from $\mathbf{F} = \mathbf{F}(\mathbf{x}, t)$ for all particles
 - $3N$ functions corresponding to $3N$ coordinates

- Forget the force. Introduce something else

- Lagrangian: $L = L(q, \dot{q})$

- Lagrange's Equation

Coordinate q and its time derivative

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}} \right) - \frac{\partial L}{\partial q} = 0$$

Everything about this system is embodied in a **scalar** function L

- Lagrangian does not depend on a coordinate system
 - Switching to a different set of coordinates is a snap

Hamilton's Principle

- **Hamilton's Principle** derives Lagrange's Equation from a simple rule:

The time integral of L is stationary for the path taken by an actual physical system

$$\delta \int_1^2 L dt = 0$$

Rather weird statement...

- **Newton's Laws were found by induction**
 - “It is so because it agrees with many observations”
 - *Deriving* them from a principle means knowing *why it is so*
 - Not quite that dramatic, but it does suggest deeper reason
 - Eventually connected to **Feynman's path integral**
- Besides, calculus of variations is a useful technique

Hamiltonian Formulation

- **Hamilton Equation**

- (p, q) are canonical variables
- H is a function called Hamiltonian

$$\dot{q} = \frac{\partial H}{\partial p}, \quad -\dot{p} = \frac{\partial H}{\partial q}$$

- Canonical variables $\sim$ position and momentum

- Aren't they related by $p = mv$?

- **Position and momentum as independent variables**

- Allows wider range of variable transformations than Lagrangian formulation
 - Formalism is clean, symmetric and cool
 - Eerie similarity to what QM does with the uncertainty principle

Success out of a Failure

- Quest for tools that solve the 3-body problem failed
 - Unless you count invention of computers
- Byproducts (Lagrangians and Hamiltonians) turned out to be the cornerstones of Quantum Mechanics
 - Development of QM was guided by analogies to Lagrangian and Hamiltonian formulations
 - Pioneers of QM grew up with Classical Mechanics
- Classical Mechanics is the missing link between Newton and Schrödinger
 - It allows you to fully appreciate QM

What We Will Study

- Lagrange's Equations, Hamilton's Principle
 - Central force problem
 - Rigid body motion
 - Oscillation
 - Extension to special relativity
- ~~Hamilton Equation, Canonical transformations~~
 - ~~Hamilton Jacobi Equation~~
- ~~Advanced stuff~~
 - ~~Classical chaos?~~
 - ~~Perturbation theory?~~
 - ~~Field theory?~~

Mechanics



Lecture 1

Elementary Principles

(Goldstein Chapter 1)

Goals for Today

- Review basic principles of Newtonian Mechanics
 - Very quickly so that you don't fall asleep
- Discuss motion of a single particle
 - Define standard notations and usages
 - Momenta, conservation laws, kinetic & potential energies
- You (should) already know all this

Single Particle

- Particle = object with insignificant size

- Electron in a CRT
- Baseball thrown by a pitcher
- Earth orbiting the Sun

- It has mass m , and location $\mathbf{r}$

- Velocity $\mathbf{v} = \frac{d\mathbf{r}}{dt} = \dot{\mathbf{r}}$

- Linear momentum $\mathbf{p} = m\mathbf{v} = m\dot{\mathbf{r}}$

- Newton's 2nd law of motion $\Rightarrow \mathbf{F} = \frac{d\mathbf{p}}{dt} = \dot{\mathbf{p}} = m\ddot{\mathbf{r}}$

Inertial System

- The origin O of $\mathbf{r}$ is somewhat arbitrary
 - A choice of origin $\rightarrow$ a reference frame
- Inertial system = a reference frame in which $\mathbf{F} = \dot{\mathbf{p}}$ holds
 - Newton's 2nd Law should be stated more precisely as

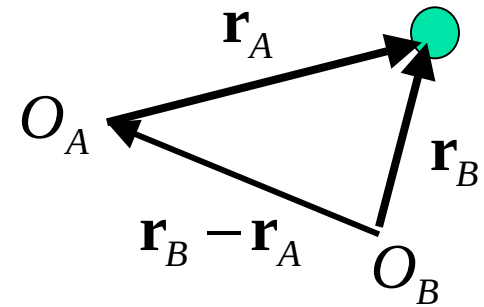
There exist reference frames in which the time derivative of the linear momentum equals to the force
 - And there are infinite number of such frames

Inertial Systems

- Consider two inertial systems A and B

- A particle is at $\mathbf{r}_A$ in A , $\mathbf{r}_B$ in B
- Origin of A is at $\mathbf{r}_B - \mathbf{r}_A$ in B

$$\begin{aligned}\mathbf{F} = m\ddot{\mathbf{r}}_A &= m\ddot{\mathbf{r}}_B \rightarrow \ddot{\mathbf{r}}_B - \ddot{\mathbf{r}}_A = 0 \\ &\rightarrow \dot{\mathbf{r}}_B - \dot{\mathbf{r}}_A = \text{const}\end{aligned}$$



Any two inertial systems are moving relative to each other at a constant velocity

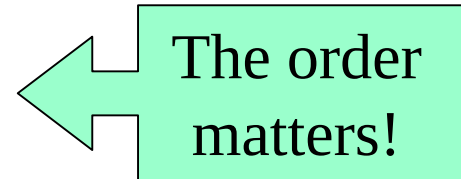
- Equivalence of such systems was pointed out by Galileo
 - Hence the name **Galilean system**

Angular Momentum

- Define

- Angular momentum $\mathbf{L} = \mathbf{r} \times \mathbf{p}$

- Moment of force (= torque) $\mathbf{N} = \mathbf{r} \times \mathbf{F}$



The order matters!

- From $\mathbf{F} = \dot{\mathbf{p}}$ one can deduce $\mathbf{N} = \dot{\mathbf{L}}$

- Too easy to write down here

- Subtlety: the definitions depend on the origin O

- Because $\mathbf{r}$ is defined from O

- The equation $\mathbf{N} = \dot{\mathbf{L}}$ holds for any origin

Momentum Conservation

- Two conservation theorems follow

- From $\mathbf{F} = \dot{\mathbf{p}}$

If the total force $\mathbf{F}$ is zero,
the linear momentum $\mathbf{p}$ is conserved

- From $\mathbf{N} = \dot{\mathbf{L}}$

If the total torque $\mathbf{N}$ is zero,
the angular momentum $\mathbf{L}$ is conserved

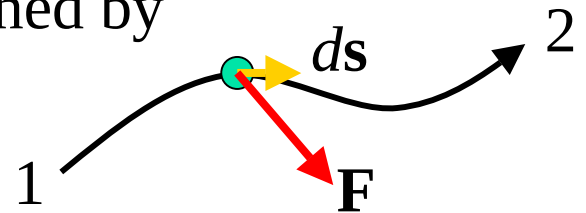
This is *really* getting too easy...

Work by External Force

- Particle moves from point 1 to 2 under force $\mathbf{F}$

- Work W_{12} done by the force $\mathbf{F}$ is defined by

$$W_{12} = \int_1^2 \mathbf{F} \cdot d\mathbf{s}$$



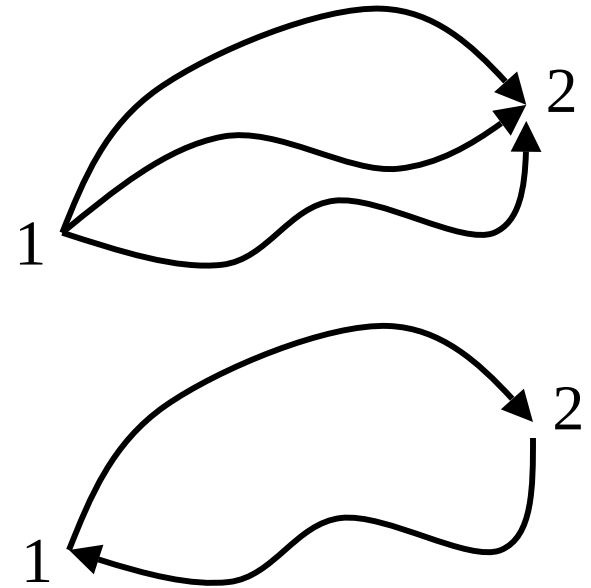
- One can define the kinetic energy $T \equiv \frac{mv^2}{2}$
- Then derive $W_{12} = T_2 - T_1$

Work done equals to the change in the kinetic energy

Conservative Force

- If W_{12} is the same for any possible path from 1 to 2, the force $\mathbf{F}$ is **conservative**
 - W_{12} depends only on the end points, not on the path
- Equivalently, if you make a closed loop, the total work is zero

$$\oint \mathbf{F} \cdot d\mathbf{s} = \int_1^2 \mathbf{F} \cdot d\mathbf{s} + \int_2^1 \mathbf{F} \cdot d\mathbf{s} = 0$$



Potential Energy

- $\mathbf{F}$ is conservative $\Leftrightarrow \mathbf{F}$ is expressed by $\mathbf{F} = -\nabla V(\mathbf{r})$
 - V is the potential energy
- Work W_{12} is then expressed by $W_{12} = \int_1^2 \mathbf{F} \cdot d\mathbf{s} = V_1 - V_2$
 - Which was equal to $T_2 - T_1$



$$T_1 + V_1 = T_2 + V_2$$

If the force is conservative, the total energy $T + V$ is conserved

Energy Conservation Theorem

Summary

- Reviewed basic principles of Newtonian Mechanics
 - Define standard notations and usages
 - Momenta, conservation laws, kinetic & potential energies
- I hope everything looked familiar, if boring
 - It will get better from here ☺
- Next: multi-particle system & constraints